UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 15

Content.

1. Applications of McDiarmid's Inequality
   (a) MAXCUT
   (b) Graph coloring
2. Efron–Stein inequality

1 Application of McDiarmid's Inequality

Reminder (McDiarmid's Inequality): Suppose $X_1, \ldots, X_n$ are independent and $g : \mathbb{R}^n \to \mathbb{R}$ satisfies the *bounded differences* property, namely that

$$|g(x_1, \ldots, x_i, \ldots, x_n) - g(x_1, \ldots, x_i', \ldots, x_n)| \leq d_i$$

for all i and for all $x_1, \ldots, x_{i-1}, x_i, x_i', x_{i+1}, \ldots, x_n$. Then,

$$\mathbb{P}[|g(X_1, \ldots, X_n) - Eg(X_1, \ldots, X_n)| \geq t] \leq 2\exp(-\frac{t^2}{2\sum_{i=1}^n d_i^2})$$

1.1 MAXCUT

MAXCUT: Given a graph $G = (V, E)$, a cut is defined by a mapping $\sigma : V \to \{0, 1\}$, with cut value

$$\mathrm{MC}(\sigma) = |\{(i, j) \in E : \sigma(i) \neq \sigma(j)\}|.$$

Two facts:

1. $\mathrm{MC}(\sigma) \leq |E|$.

2. If $\sigma(i)$ are i.i.d. $\mathrm{Bernoulli}(1/2)$, then

$$\mathbb{E}[\mathrm{MC}(\sigma)] = \frac{|E|}{2}.$$

Therefore, there exists a cut $\sigma : V \to \{0,1\}$ with cut value $\mathrm{MC}(\sigma) \geq \frac{|E|}{2}$.

Define the max cut as

$$\mathrm{MC}(G) := \max_{\sigma: V \to \{0,1\}} \mathrm{MC}(\sigma).$$

Then

$$\frac{|E|}{2} \leq \mathrm{MC}(G) \leq |E|.$$

It is NP-hard to find the cut maximizing above.

Let's consider $G(n, \frac{d}{n})$, where each edge appears independently with probability $\frac{d}{n}$.

Q: what is the 'value' of $MC(G(n, \frac{d}{n}))$?

With high probability, $G(n, \frac{d}{n})$ has $\binom{n}{2}\frac{d}{n} = \frac{dn}{2}(1 + o_n(1))$ edges:

$$\frac{d}{4} \leq \frac{1}{n}\mathrm{MC}(G(n, \frac{d}{n})) \leq \frac{d}{2}$$

So, $\mathrm{MC}(G(n, \frac{d}{n})) = \Theta(n)$ with high probability.
Want: $\frac{1}{n}\mathrm{MC}(G(n, \frac{d}{n}))$
Strategy: Show concentration around its mean and study mean instead

Open Problem. The value of

$$\lim_{n\to\infty} \frac{1}{n}\mathbb{E}[\mathrm{MC}(G(n, \frac{d}{n}))]$$

is unknown. The existence of the limit was shown in 2010 using the interpolation method by Bayati-Gamarnik-Tetali.

Applying McDiarmid's Inequality: Let

$$\mathrm{MC} := \mathrm{MC}(E_1, \ldots, E_{\binom{n}{2}})$$

Namely, $\mathrm{MC}(G(n, \frac{d}{n}))$ is a function of $\binom{n}{2}$ edges that are i.i.d. Bern($\frac{d}{n}$). We have that changing one edge affects the cut value by at most 1:

$$|\mathrm{MC}(E_1, \ldots, E_i, \ldots, E_{\binom{n}{2}}) - \mathrm{MC}(E_1, \ldots, E_i', \ldots, E_{\binom{n}{2}})| \leq 1.$$

Thus, by McDiarmid's inequality,

$$\mathbb{P}\left(|\mathrm{MC}(G(n,\frac{d}{n})) - \mathbb{E}[\mathrm{MC}(G(n,\frac{d}{n}))]| \geq t\right) \leq 2\exp\left(-\frac{t^2}{2\binom{n}{2}}\right) = 2\exp\left(-\frac{t^2}{n(n-1)}\right).$$

If we plug $t = \varepsilon\,\mathbb{E}[\mathrm{MC}(G(n,\frac{d}{n}))]$, then this bound becomes

$$2\exp\left(-\frac{\varepsilon^2\mathbb{E}[\mathrm{MC}(G(n,\frac{d}{n}))]^2}{n(n-1)}\right) = 2\exp(-\Theta(1))$$

and McDiarmid's fails.

Fix: we introduce a slightly different model: $G(n,m)$, where one chooses m edges uniformly at random among the set of all $\binom{n}{2}$ edges.

$$\Omega = \{(i,j) : 1 \leq i < j \leq n\}$$

and $G(n,m)$ are defined by m i.i.d. random variables $E_1, \ldots, E_m \sim \mathrm{Uniform}(\Omega)$.

Remark. *Parallel edges $i-j$ are negligible.*

Fact: The graphs $G(n, m = \frac{dn}{2})$ and $G(n, p = \frac{d}{n})$ are "contiguous", i.e. they behave similarly.

Concentration of $\mathrm{MC}(G(n, m = \frac{dn}{2}))$ On $G(n,\frac{dn}{2})$, MC satisfies

$$|\mathrm{MC}(E_1,\ldots,E_i,\ldots,E_m) - \mathrm{MC}(E_1,\ldots,E_i',\ldots,E_m)| \leq 1.$$

Therefore, by McDiarmid's inequality,

$$\mathbb{P}\left(|\mathrm{MC} - \mathbb{E}\mathrm{MC}| \geq t\right) \leq 2\exp\left(-\frac{t^2}{dn}\right)$$

If $t = \Theta(n)$, we get

$$\mathbb{P}\left(|\mathrm{MC} - \mathbb{E}[\mathrm{MC}]| \geq t\right) \leq \exp(-\Theta(n)).$$

Since $\mathbb{E}[\mathrm{MC}] = \Theta(n)$,

- Deviating from $\mathbb{E}[\mathrm{MC}]$ by order of $\Theta(\mathbb{E}[\mathrm{MC}])$ is $\exp(-\Theta(n))$. This is a large deviation.
- If $t = \Theta(\sqrt{n})$, $t = r\sqrt{n}$,

  $$\mathbb{P}\left(|\mathrm{MC} - \mathbb{E}[\mathrm{MC}]| \geq t\right) \leq 2\exp(-\frac{r^2}{d})$$

  This is sub-Gaussian tails. We have $\mathrm{MC} = \mathbb{E}\mathrm{MC} + \Theta(\sqrt{n})$, this is meaningful, as $\mathbb{E}\mathrm{MC} = \Theta(n)$.

1.2 Graph coloring

Given $G = (V, E)$, a coloring is a mapping $\phi : V \to \{1, 2, \ldots, q\}$ such that if $(i, j) \in E$, then $\phi(i) \neq \phi(j)$.

Definition 1 (Chromatic Number)**.** *$\chi(G)$, the chromatic number of G, is the smallest q for which a valid q-coloring exists.*

Example :

- If $G = K_n$, n-clique, then $\chi(G) = n$
- If G is bipartite, then $\chi(G) = 2$
- $\chi(G) \geq 2$, unless G is empty graph

Random Graphs: $G(n, p)$, each edge appears with probability $p > 0$

Theorem 1 (Bolloba's 89)**.**

$$\chi(G(n,p)) = \frac{n}{2\log_{1/p} n}\,(1 + o(1)).$$

and $\mathbb{E}[\chi(G(n,p))] = \Theta\left(\frac{n}{\log n}\right)$.

Question: How about concentration?

$$\chi = \chi(E_1, E_2, \ldots, E_{\binom{n}{2}})$$
$$\left|\chi(E_1, \ldots, E_i, \ldots, E_{\binom{n}{2}}) - \chi(E_1, \ldots, E_i', \ldots, E_{\binom{n}{2}})\right| \leq 1$$
$$\mathbb{P}\left(|\chi(G(n,p)) - \mathbb{E}[\chi(G(n,p))]| \geq t\right) \leq 2\exp\left(-\frac{t^2}{2\binom{n}{2}}\right).$$

If $t = \Theta\big(\mathbb{E}[\chi(G(n,p))]\big) = \Theta\big(\frac{n}{\log n}\big)$, then the bound becomes

$$\exp\left(-\frac{n^2/(\log n)^2}{2\binom{n}{2}}\right) \sim \exp(-\frac{1}{(\log n)^2}) \to 1$$

Vertex Exposure Martingale (Shamir–Spencer, '89) Let E_{ij} denote the status of the edge between i and j, and define

$$\bar{E}_i := \{E_{ij} : 1 \leq j < i\} \quad (\bar{E}_1 = \varnothing)$$

the collection of edges connecting vertex i to all previous vertices.

Let

$$X_i = \mathbb{E}[\chi(G(n,p)) \mid \bar{E}_1, \ldots, \bar{E}_i],$$

where $\bar{E}_1, \ldots, \bar{E}_i$ record the edges between the first i vertices. Then, X_i is a Doob martingale.

Exercise 1.

$$|X_{i+1} - X_i| \leq 1, \quad \forall i$$

Azuma's Inequality asserts that for

$$X_n - X_0 = \chi(G(n,p)) - \mathbb{E}[\chi(G(n,p))]$$

we have

$$\mathbb{P}\left(|X_n - X_0| \geq t\right) \leq 2\exp\left(-\frac{t^2}{n}\right).$$

If $t = \Theta\left(\frac{n}{\log n}\right)$, then the bound is $\exp\left(-\Theta\left(\frac{n}{(\log n)^2}\right)\right)$ (almost exponential)

2 Efron-Stein inequality

Setup: $f : \mathbb{R}^n \to \mathbb{R}$ and let $X_1, \ldots, X_n$ be independent random variables.
Goal: Bound $\mathrm{var}(f(X))$, where $X = (X_1, \ldots, X_n)$
Notes:

1. The inequality does not provide exponential concentration.
2. It does not require the bounded difference property.

Theorem 2 (Efron-Stein)**.**

$$\mathrm{var}(f(X)) \leq \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}\left[\left(f(X_1, \ldots, X_i, \ldots, X_n) - f(X_1, \ldots, X_i', \ldots, X_n)\right)^2\right].$$

where $X_i' \perp X_i$ is an independent copy of X_i.

Lemma 1. *If* $X \overset{i.i.d.}{\sim} X'$*, then* $\mathrm{var}(X) = \frac{1}{2}\mathbb{E}[(X - X')^2]$

Proof.

$$\frac{1}{2}\mathbb{E}[(X - X')^2] = \frac{1}{2}\mathbb{E}[X^2 - 2XX' + X'^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathrm{var}(X).$$

□

Proof Idea: Define the Doob martingale:

$$M_i := \mathbb{E}[f(X) \mid X_1, \ldots, X_i],$$

- Without loss of generality, assume $\mathbb{E}[f(X)] = 0$
- Next lecture: let $D_i = M_i - M_{i-1}$, control $\mathbb{E}[(\sum_i D_i)^2]$